

- Spatial Intelligence & Multiple-View Geometry

Feature Matching in Computer Vision

Unlocking Spatial Intelligence Across Multiple Views: From Fundamental Geometry to AI-Powered Feature Alignment.

SPEAKER 1

Pavan Kumar S

Foundations & Detection

SPEAKER 2

Keerthana

Descriptors & Matching

SPEAKER 3

Thanushree

Learned Models & SfM

9 Slides

Triad Team Delivery

What is Feature Matching?

The primary mechanism for establishing spatial correspondences across separate vantage points

CORE OBJECTIVE

Establishing Mathematical Ground Truth

To identify and establish reliable spatial correspondences between identical physical surface points depicted across disparate viewpoints, sensor modalities, time offsets, and illumination conditions.

Geometric Formulation:

Given point $\mathbf{x} \in \mathbf{I}_1$, determine corresponding point $\mathbf{x}' \in \mathbf{I}_2$ such that both image coordinates originate from identical 3D world coordinate $\mathbf{X} \in \mathbb{R}^3$.

Foundational for: Visual Odometry, Structure from Motion, and 3D Perception.

ENVIRONMENTAL OBSTACLES

Three Inherent Vision Challenges

1 Perspective & Affine Distortions
Camera rotation and translation generate severe non-linear surface foreshortening.

2 Illumination & Photometric Shifts
Diurnal solar shifts, specular reflection, and varying sensor exposure ranges.

3 Scale Variations & Dynamic Occlusions
Target distance disparities alter resolution; transient foreground obstacles block sightlines.

CANONICAL 4-STAGE PIPELINE:

1. Detection → 2. Description → 3. Matching → 4. Outlier Rejection

Detecting Interest Points

Extracting distinct, spatially repeatable anchors across multi-scale image spaces

CORNER DETECTORS

Harris / Shi-Tomasi

Multidirectional Shift

Identifies points where image gradient variation is substantial across two orthogonal directions simultaneously.

Second Moment Matrix M :

$$R = \det(M) - k(\text{trace}(M))^2$$

Eigenvalues (λ_1, λ_2) must both be large and positive.

Optimal for: High-contrast rectilinear architectural scenes.

BLOB DETECTORS

DoG / Hessian

Scale-Space Extrema

Isolates uniform brightness patches by evaluating difference-of-Gaussians across continuous scale octaves.

Gaussian Scale Space:

$$D(x, y, \sigma) = (G(k\sigma) - G(\sigma)) * I$$

Scale invariance achieved through 3D continuous extrema testing.

Optimal for: Multi-scale satellite and aerial registration.

REAL-TIME POINTING

FAST Algorithm

Segment Test Heuristic

Building Feature Descriptors

Converting local spatial neighborhoods into robust, invariant mathematical fingerprints

CONTINUOUS VECTOR

128-D Float32

Floating-Point Descriptors (SIFT / SURF)

Constructs localized gradient orientation histograms computed over a canonical 16×16 pixel grid centered on the detected interest point.

SIFT Vector Construction:

16 sub-regions (4×4) × 8 orientation bins = **128 floating-point values**. Normalized with L2 norm to suppress illumination gain.

STRENGTHS

Scale & Rotation Invariant

TRADE-OFF

Heavy Memory & CPU Cost

Primary Application: High-accuracy photogrammetry and geospatial mapping.

BINARY BITSTRING

256-Bit Bitstring

Binary Descriptors (ORB / BRIEF / BRISK)

Encodes relative pairwise pixel intensity tests into compact bit strings, steered by the intensity centroid to achieve rotation invariance.

Point Comparison Function τ :

$\tau(p; x, y) = 1$ if $I(p(x)) < I(p(y))$, else 0

256 tests assembled directly into a 32-byte integer array.

STRENGTHS

Native POPCNT Acceleration

TRADE-OFF

Reduced Scale Dynamics

Matching Mechanics & Distance Metrics

Pairwise proximity evaluation and ambiguity pruning in multidimensional metric spaces

METRICS

Distance Metrics

Determined strictly by descriptor vector representation:

Euclidean (L2 Norm)

Applied to Float vectors (SIFT). Computes vector difference in $\mathbb{R}^{128}$.

Hamming Distance

Bitwise XOR followed by population count (POPCNT). 1 CPU instruction.

Metrics must strictly satisfy the triangle inequality.

INDEX ENGINE

Search Strategies

Algorithmic nearest neighbor resolution:

Brute-Force Matcher

Exhaustive $O(N \times M)$ comparison. Guaranteed global optimum, high compute cost.

FLANN Tree Search

Randomized KD-Trees & hierarchical k-means for approximate nearest neighbors in $O(\log N)$.

FLANN offers ~10x acceleration on dense datasets.

AMBIGUITY FILTER

Lowe's Ratio Test

Pruning repetitive textures & percentual ambiguity:

Outlier Rejection & Geometrical Verification

Eliminating spurious correspondences through spatial constraints and consensus fitting

PRE-FILTERING STATE

High Outlier Contamination

Raw nearest neighbor matching produces massive noise caused by dynamic scene objects, repetitive patterns, and optical sensor noise.

Typical Inlier Ratio:	15% – 35%
Outlier Contamination:	65% – 85%

Unfiltered matches lead to catastrophic 3D pose failure.

RANSAC Algorithm Loop

ITERATIVE CONSENSUS

Step 1

Sample minimal subset (4 or 8 pts)

Step 2

Hypothesize geometric model matrix

Step 3

Count inliers within error threshold ϵ

Step 4

Refine model over maximum consensus set

Homography Matrix ($H \in \mathbb{R}^{3 \times 3}$)

For strictly planar scenes or pure camera rotation. Requires 4 point correspondences.

$x' \sim H x$

Fundamental / Essential (F, E)

Enforces epipolar geometry for unconstrained 3D camera translation. Requires 8 points.

$x'^T F x = 0$

Guaranteed Filtering Efficacy: Filters up to 80%+ raw false positives into sub-pixel accurate spatial alignments

Learned Matching Architectures

Replacing handcrafted heuristics with differentiable graph neural networks and deep priors

FRONT-END EXTRACTOR

MagicLeap / CVPR

SuperPoint: Self-Supervised Joint Extraction

A fully convolutional network architecture operating on full-sized images to compute interest points and dense descriptors in a single forward pass.

Shared VGG Encoder Backbone

Single shared representation branches into a Keypoint Detector Head (sub-pixel heatmap) and Dense Descriptor Head (semi-dense 256-D vectors).

Homographic Adaptation

Self-supervised multi-scale warping pipeline completely eliminates reliance on human keypoint labeling.

Output: Highly repeatable keypoints under extreme blur and lighting swings.

GRAPH MATCHER · GNN

Attentional GNN / CVPR

SuperGlue: Attentional Graph Matching

Solves data association and outlier rejection simultaneously using a Graph Neural Network with Self- and Cross-Attention layers.

Self & Cross Attention Layers

Self-attention propagates spatial context within the same image; cross-attention exchanges visual semantics across views, mirroring human gaze.

Optimal Transport (Sinkhorn Algorithm)

Formulates assignment as an optimal transport problem with an explicit dustbin mechanism for unmatched occlusions.

Result: End-to-end differentiable matching without heuristic RANSAC tuning.

Algorithm Selection & Benchmark Guide

Empirical trade-offs across computational budgets, precision thresholds, and platform hardware

Algorithm	Vector Type	Inference Latency	Scale Robustness	Illumination Invariance	Primary Target Application
SIFT	128-D Float32	~70 ms (CPU)	Very High	Moderate	High-Precision Geospatial & SfM
ORB	256-bit Binary	< 3 ms (Ultra-Fast)	Moderate	Low – Moderate	Embedded SLAM & Edge Robotics
SuperPoint + SuperGlue	Learned Attention	12 ms (GPU)	Extreme	State-of-the-Art	Day/Night Autonomy & AR Headsets

RESOURCE CONSTRAINED?

Deploy ORB / FAST

Guarantees 30+ FPS on edge ARM architectures.

PHOTOGRAMMETRY QUALITY?

Deploy SIFT + RANSAC

Proven metric repeatability across survey scales.

CHALLENGING LIGHTING?

Deploy SuperPoint + SuperGlue

Resilient to low-texture and low-lux night scenes.

Real-World Applications & Conclusion

Empowering spatial intelligence across industries and commercial devices

01 · PHOTOGRAPHY

Image Stitching

Computes planar homographies to seamlessly merge multiple overlapping camera frames into high-resolution panoramas without seam artifacts.

Mobile Devices & Satellites

02 · ROBOTICS

Visual SLAM

Simultaneous Localization and Mapping for autonomous vehicles and UAVs, estimating real-time 6-DoF trajectories in GPS-denied environments.

Autonomous Systems

03 · PHOTOGRAMMETRY

3D Reconstruction

Structure from Motion (SfM) pipelines extracting bundle-adjusted keypoints to generate millimeter-scale digital twins of architecture.

Digital Twins & Heritage

04 · SPATIAL COMPUTING

Augmented Reality

Locks virtual interactive holographic assets firmly onto physical surfaces via real-time planar tracking and visual relocalization.

AR Glasses & Spatial OS

EXECUTIVE SYNTHESIS

“Feature matching seamlessly bridges 2D camera pixels to 3D physical spatial comprehension.”

Thank you for your attention

Questions & Discussion